
Reshaping Reasoning in LLMs: A Theoretical Analysis of RL Training Dynamics through Pattern Selection

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Abstract

1 Although reinforcement learning (RL) has proven highly effective at improving
2 the reasoning capabilities of large language models (LLMs), the optimization
3 dynamics that govern RL fine-tuning of LLMs remain insufficiently characterized.
4 In this work, we account for the RL training procedure by combining a controlled
5 empirical analysis with a formal dynamical model. We first conduct a two-level
6 diagnostic study of the training trajectory—at the granularity of reasoning patterns
7 and of individual tokens—and establish two quantitative regularities: the per-pattern
8 success rate stays approximately invariant over training, whereas RL concentrates
9 its updates on a sparse subset of decision-critical tokens, and it is this reallocation of
10 probability mass that reshapes the distribution over reasoning patterns and thereby
11 determines model accuracy. Building on these measurements, we construct a
12 theoretical framework that characterizes the training dynamics of RL under two
13 canonical reward specifications: reinforcement learning with verifiable rewards
14 (RLVR) and reinforcement learning with internal feedback (RLIF). For RLVR, we
15 analyze the gradient-flow dynamics in two limiting regimes—one in which the
16 policy converges efficiently to the optimal reasoning strategy, and one in which
17 convergence is obstructed—and show that the reasoning quality of the base model is
18 the decisive factor separating the two regimes. For RLIF, we prove that the internal
19 reward raises accuracy at the start of training but can drive the policy toward a
20 lower-accuracy fixed point as optimization continues. A suite of experiments
21 corroborates these predictions, advancing both the theoretical understanding and
22 the practical deployment of RL for LLM enhancement.

23 1 Introduction

24 State-of-the-art reasoning models—including Gemini2.5 [8], Qwen3 [40], and DeepSeek-R1 [14]—
25 now attain strong performance on tasks that demand multi-step logical inference, most notably
26 mathematical problem solving [30, 44, 41] and program synthesis [49, 40, 8]. Reinforcement
27 learning (RL) is a principal driver of these gains, functioning as the post-training stage that lifts model
28 capability beyond what pretraining and supervised fine-tuning alone achieve.

29 This empirical success has motivated a body of work aimed at identifying the mechanism by which RL
30 improves LLMs. Measuring $\text{pass}@k$ as a function of k , Yue et al. [42] report that RL post-training
31 rarely expands the set of solvable problems relative to the base model, indicating that RL does
32 not instantiate qualitatively new reasoning behaviors but rather re-weights existing ones. From an
33 information-theoretic standpoint, Cui et al. [9], Zhang et al. [45] prove that RL objectives contract
34 the entropy of the policy, and Cui et al. [9] additionally document an empirical correlation between
35 accuracy and policy entropy. At the token level, Wang et al. [33], Huan et al. [17] show that the
36 effective updates are confined to a sparse subset of high-leverage tokens. Notwithstanding these

contributions, a precise account of the RL *training dynamics*—how the policy evolves over the course of optimization—is still lacking at both the empirical and the theoretical level.

To characterize the training process, we begin with a controlled diagnostic study conducted at two levels of granularity—the reasoning-pattern level and the token level—and tracked across the full optimization trajectory. Concretely, we (i) measure the shifts in next-token probability ranks induced by training, and (ii) apply both an LLM-based classifier and rule-based extractors to segment model responses into reasoning patterns, then quantify each pattern’s frequency and success rate as training proceeds. Relative to prior analyses [17, 42], this protocol yields sharper and more direct evidence that RL updates concentrate on a sparse subset of critical tokens and, through them, *reshape the distribution over reasoning patterns* so as to modulate accuracy. We further find that the intrinsic per-pattern success rate is approximately invariant under training. These measurements motivate the mathematical framework we develop to formalize the RL training process.

Grounded in these findings, we formalize reasoning as a two-stage generative process, $q \rightarrow r \rightarrow a$, that factorizes the policy into (1) selection of a reasoning pattern conditioned on the question, $\pi(r|q)$, and (2) answer generation conditioned on the selected pattern, $\pi(a|r, q)$. Within this factorization we analyze the training dynamics of RL under two canonical reward specifications: reinforcement learning with verifiable rewards (RLVR) [14, 30] and reinforcement learning with internal feedback (RLIF) [47, 4]. For RLVR, we prove that the policy converges to the reasoning pattern r^* of maximal success rate, in exact agreement with our measurements, and we delineate two convergence regimes: base models of high reasoning quality converge rapidly to r^* , whereas lower-quality base models pass through an entanglement stage in which convergence is severely slowed. For RLIF, we identify the mechanism responsible for the early-training accuracy gain and prove that continued optimization can drive the policy to a fixed point whose accuracy falls below that of the base model, consistent with the observed instability. Additional case studies and numerical simulations validate the analysis.

The main contributions of this paper are highlighted as follows:

- We carry out a systematic diagnostic study of the RL training trajectory at the reasoning-pattern and token levels. Combining measurements of next-token rank shifts with LLM-based and rule-based pattern extraction, we obtain sharper and more direct evidence of the training dynamics than prior work [17, 42]: RL confines its effective updates to a sparse subset of critical tokens and thereby *reshapes the distribution over reasoning patterns* to modulate accuracy, while the intrinsic per-pattern success rate remains approximately invariant throughout training.
- We formalize reasoning as a two-stage process $q \rightarrow r \rightarrow a$ (question–reason–answer) and derive the training dynamics of two canonical RL objectives within this model. For RLVR, we prove convergence to the maximal-success-rate reasoning pattern and separate two regimes—efficient convergence for high-quality base models versus an entanglement-stage slowdown for lower-quality ones. For RLIF, we prove that accuracy increases at initialization and characterize the conditions under which continued optimization converges to a fixed point with accuracy below that of the base model.
- We validate the analysis with controlled case studies and numerical simulations that reproduce each predicted regime. The framework closes the gap between the empirical observations and a formal account of RL training dynamics, and yields actionable guidance for RL-based LLM post-training.

2 Related Works

Reinforcement Learning for LLMs Reinforcement learning has become a standard post-training procedure for large language models (LLMs), applied both to aligning model outputs with human preferences [25, 48] and to eliciting competitive performance on mathematical and programming benchmarks [30, 20, 22]. The reward signal is the defining component of any RL pipeline; classical formulations instantiate it as a learned reward model fit to human-annotated preference data [26, 25, 2], which incurs substantial annotation and training cost. Recent work replaces this component with cheaper, automatically computable signals—verifiable outcome rewards, as in reinforcement learning with verifiable rewards (RLVR) [30, 40, 14], and intrinsic model-derived rewards, as in reinforcement learning with internal feedback (RLIF) [47, 4]. In parallel, the optimization procedure itself has been refined through a succession of policy-gradient algorithms, including GRPO [30], DAPO [41], and VAPO [43].

Theoretical Analyses and Mechanism Interpretation of RL for LLMs. The empirical success of RL for LLMs has prompted complementary efforts to formalize and interpret its behavior. Early theoretical work operated at a coarse granularity, addressing reward-design principles [28, 18, 38] and the construction of new optimization algorithms [37, 11, 21]. As the field moved toward the simpler reward structures of RLVR and RLIF, subsequent studies sought to isolate the source of their effectiveness. Shao et al. [29] attribute the gains to the reactivation of latent pretraining skills under spurious rewards; Agarwal et al. [4] account for them through entropy minimization; and Gandhi et al. [13] localize them to specific cognitive behaviors, such as backtracking. Zhao et al. [46] show that RL fine-tuning amplifies behaviors already present at pretraining, collapsing generation onto a dominant output format—a phenomenon that Wang et al. [34], Cui et al. [10] further characterize as entropy collapse. In contrast to these accounts, we analyze the RL training dynamics through the lens of reasoning-pattern selection and support the analysis with a formal framework and provable convergence results.

3 Preliminaries

Reinforcement Learning for LLMs Let π_θ denote an autoregressive language model with parameter vector θ , treated as the policy to be optimized. Conditioned on an input question $\mathbf{x} = (x_0, x_1, \dots, x_n)$, the policy samples a response $\mathbf{y} = (y_0, y_1, \dots, y_m)$. The learning problem is the KL-regularized reward-maximization objective

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_\theta(\cdot | \mathbf{x})} [r_\phi(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}}[\pi_\theta(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (3.1)$$

where π_{ref} denotes the fixed reference (base) policy and the coefficient $\beta \geq 0$ weights the reverse-KL penalty $\mathbb{D}_{\text{KL}}[\pi_\theta \parallel \pi_{\text{ref}}]$, which regularizes the trajectory of π_θ toward π_{ref} and bounds its deviation from initialization.

The reward $r_\phi(\mathbf{x}, \mathbf{y})$ admits several instantiations, ranging from a learned reward model [25] to a rule-based scoring function [30]. We restrict attention to two verifier-free specifications: verifiable rewards (RLVR) [14, 30] and internal-feedback rewards (RLIF) [47, 4].

Under RLVR, the reward is the indicator that the generated answer $\mathbf{y}$ agrees with the ground-truth answer for $\mathbf{x}$, i.e.,

$$r_\phi(\mathbf{x}, \mathbf{y}) = \begin{cases} 1 & \text{if } \mathbf{y} = \text{the ground truth of } \mathbf{x}, \\ 0 & \text{otherwise.} \end{cases} \quad (3.2)$$

In contrast, RLIF [47, 4] derives its reward exclusively from intrinsic quantities of the policy, dispensing with any external verifier. Following Zhao et al. [47], we take the RLIF reward to be the token-averaged negative KL divergence between the uniform distribution U over the vocabulary $\mathcal{V}$ and the model’s next-token distribution, i.e.,

$$r_\phi(\mathbf{x}, \mathbf{y}) = \frac{1}{|\mathbf{y}|} \sum_{i=1}^{|\mathbf{y}|} \mathbb{D}_{\text{KL}}(U \parallel \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})) = -\frac{1}{|\mathbf{y}| \cdot |\mathcal{V}|} \sum_{i=1}^{|\mathbf{y}|} \sum_{j=1}^{|\mathcal{V}|} \log(|\mathcal{V}| \cdot \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})). \quad (3.3)$$

Our objective in this paper is to characterize the optimization dynamics induced by both the RLVR and RLIF reward specifications, through matched empirical measurement and formal analysis.

4 Experimental Exploration for RL Training

To characterize the RL training process, we conduct a controlled empirical study spanning the two reward specifications and organized as a three-level analysis. We first establish the macroscopic behavior by tracking the training-accuracy trajectory, which shows that RLVR produces monotone, saturating gains whereas RLIF is non-monotone and can regress below the base model. We then resolve these trajectories at the reasoning-pattern level, attributing the RLVR gain to a progressive reallocation of probability mass toward higher-success-rate patterns and the RLIF instability to its failure to specialize. Finally, at the token level, we identify the microscopic driver of these pattern-distribution shifts: they are produced by rank changes in a small fraction of the tokens. Relative to prior work [17, 42], this three-level protocol supplies sharper and more direct evidence for the dynamics, establishing that RL confines its effective updates to a sparse subset of critical tokens and thereby *reshapes the distribution over reasoning patterns* to modulate accuracy.

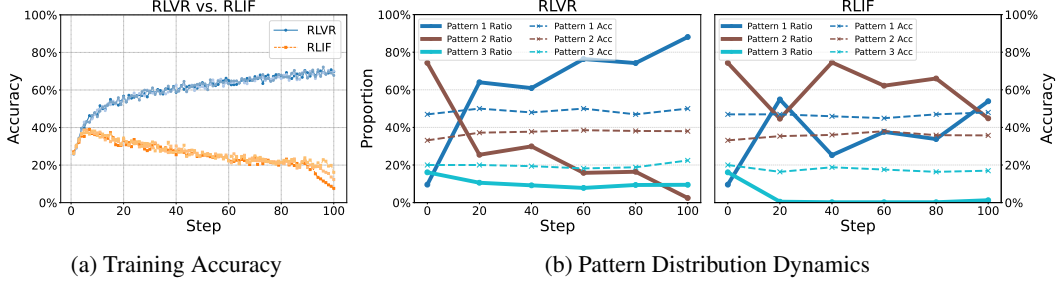


Figure 1: **(a)** Training-accuracy trajectories on the MATH dataset under RLVR and RLIF; each curve is averaged over multiple runs obtained by varying the random seed. The trajectories separate qualitatively: RLVR yields monotone, saturating gains, whereas RLIF is non-monotone, rising initially and then regressing. **(b)** Reasoning-pattern-level decomposition of the same trajectories. Solid curves report each pattern’s relative frequency among all patterns (left axis) and dotted curves its success rate (right axis). Under RLVR the policy progressively concentrates probability mass on the higher-success-rate patterns and depletes the lower-success-rate ones; under RLIF the pattern frequencies evolve erratically without specializing. Patterns are indexed in decreasing order of success rate to aid comparison.

134 4.1 Training Procedure Overview for RLVR and RLIF

135 We first characterize the full training trajectory of each reward specification. To isolate the effect of
 136 the reward, we fix the base model to Qwen2.5-3B [39] and the training set to MATH [16], holding all
 137 remaining hyperparameters identical across the two algorithms so that RLVR and RLIF differ only in
 138 the reward term.

139 The resulting accuracy curves, reported in Figure 1a, separate cleanly. RLVR induces a stable
 140 trajectory: training accuracy increases monotonically and saturates as optimization proceeds. RLIF,
 141 by contrast, is non-monotone—accuracy rises during an initial phase but subsequently declines,
 142 in some runs falling below the accuracy of the untrained base model. This qualitative divergence
 143 motivates the finer-grained, mechanism-level analysis developed in the remainder of this section.

144 4.2 Reasoning-pattern Level Analysis

145 To explain the divergent accuracy trajectories of RLVR and RLIF, we decompose them at the
 146 reasoning-pattern level. The pattern taxonomy is constructed as follows: we sample responses from
 147 the base model and prompt GPT-4o [19] to cluster them into a fixed set of categories defined by
 148 shared keywords and logical structure, emitting a textual description per category. Conditioning on
 149 this taxonomy, we then track, at successive training checkpoints, both the frequency with which each
 150 pattern is generated and its empirical success rate.

151 Concretely, we trace the joint evolution of pattern frequency and per-pattern accuracy under both
 152 objectives. Figure 1b presents this analysis on Number Theory problems drawn from MATH [16];
 153 corresponding results for additional models and datasets appear in Appendix A. Three regularities
 154 emerge:

- 155 • Under RLVR, probability mass migrates monotonically toward the higher-success-rate patterns,
 156 which accounts for the steady rise in overall accuracy.
- 157 • Under RLIF, the pattern distribution evolves erratically and does not concentrate on the more
 158 effective patterns.
- 159 • For both objectives, the success rate of each individual pattern is approximately invariant across the
 160 training trajectory.

161 Taken together, these regularities localize the RLVR–RLIF performance gap to a difference in pattern-
 162 selection dynamics rather than in the patterns themselves. We next probe the token-level mechanism
 163 that produces these distributional shifts.

4.3 Token-level Analysis

To identify the token-level mechanism underlying the pattern-distribution shifts, we adopt the rank-comparison protocol of Huan et al. [17], Yue et al. [42]. We sample base-model responses to questions from GSM8K [6], MATH [16], and AIME [7], and then, holding these responses fixed, recompute the rank of each response token under both the base and the RL-trained policy, where a token’s rank is the position of its conditional probability among all vocabulary entries given the preceding context. As summarized in Table 1, the ranks are largely preserved: in most configurations fewer than 10% of tokens change rank. Hence RL alters the conditional distribution at only a sparse set of decision-critical positions while leaving the bulk of the generated reasoning trajectory intact.

These measurements furnish direct evidence for the following characterization of the RL training dynamics: *RL primarily optimizes a sparse subset of critical tokens, thereby reshaping reasoning pattern distributions to affect model performance. Moreover, the intrinsic success rate of individual patterns remains relatively stable during training.*

Beyond their diagnostic value, these regularities supply the structural assumptions on which we base the mathematical framework and the formal training-dynamics analysis developed in the next section.

Table 1: Fraction of response tokens whose probability rank shifts after RL, reported per dataset, reward, and training step.

Task	Method	Step20	Step40	Step60	Step80	Step100
GSM8K	RLVR	5.2%	6.0%	6.6%	7.1%	7.3%
	RLIF	6.8%	8.0%	8.9%	9.5%	10.1%
MATH	RLVR	5.3%	5.8%	6.1%	6.5%	6.6%
	RLIF	6.1%	7.0%	7.7%	8.3%	8.8%
AIME24	RLVR	5.1%	5.7%	5.9%	6.1%	6.3%
	RLIF	5.6%	6.5%	7.4%	8.2%	8.6%

5 Theoretical Characterization of RL Training Dynamics

We now lift the empirical regularities of the preceding section into a formal account of the RL training process. We first specify a theoretical framework consistent with those measurements, abstracting generation as a two-stage procedure—selection of a reasoning pattern followed by answer deduction—in which RL acts primarily on the first stage. Within this framework we derive the training dynamics for each reward specification. For RLVR, we analyze the gradient-flow dynamics and isolate two limiting regimes with qualitatively distinct convergence behavior. For RLIF, we establish both the mechanism by which the internal reward raises accuracy early in training and the conditions under which continued optimization converges to a fixed point of lower accuracy than the base model.

5.1 A Theoretical Framework for Reasoning Models

Abstract Reasoning Process. We model generation as a two-stage stochastic process: given a question q , the policy (1) draws a reasoning pattern r from the candidate set $\mathcal{R} = \{r_1, r_2, \dots\}$ according to $p(r_i|q)$, and (2) emits a final answer $a \in \mathcal{A} = \{a_1, a_2, \dots\}$ conditioned on the selected pattern, with each pattern carrying its own success rate. Under this factorization, the RLIF objective (Eq. 3.3) admits the equivalent form¹

$$r_\phi(\{r, a\}, q) := - \left(\frac{1}{|\mathcal{R}|} \sum_{r_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_\theta(r_i|q)) + \frac{1}{|\mathcal{A}|} \sum_{a_j \in \mathcal{A}} \log(|\mathcal{A}| \cdot \pi_\theta(a_j|q, r)) \right) \quad (5.1)$$

Policy Parameterization. Let $\mathcal{V} = \{q\} \cup \mathcal{R} \cup \mathcal{A}$ be the vocabulary and $\mathcal{Y} = (r, a)_{r \in \mathcal{R}, a \in \mathcal{A}}$ the set of admissible output sequences. Conditioned on q , the model factorizes its distribution over $y \in \mathcal{Y}$ autoregressively as

$$(\text{general policy}) \quad \pi_\theta(y|x) = \prod_{l=1}^{|y|} \pi_\theta(y_l|x, y_{<l}) = \prod_{l=1}^{|y|} \text{softmax}(f_\theta(x, y_{<l}))_{y_l}, \quad (5.2)$$

where $f_\theta : \mathcal{V} \rightarrow \mathbb{R}^{|\mathcal{V}|}$ is the logit map parameterized by θ , the l -th token is predicted from the prefix $y_{<l}$, and the conditional distribution is the softmax $\text{softmax}(z)_v := \exp(z_v) / \sum_{v' \in \mathcal{V}} \exp(z_{v'})$ evaluated at $z = f_\theta(x, y_{<l})$.

¹For simplicity, we omit the normalizing coefficient $1/|y|$ from Eq. 3.3, which can be treated as a constant $1/2$ in our framework.

Because the logit map of a practical transformer is highly non-linear, deriving optimization guarantees for its RL dynamics remains largely intractable [3, 23]. We therefore adopt, following [27, 24, 10, 45], the *tabular policy* parameterization—the special case of Eq. 5.2 in which the conditional distribution depends only on the immediately preceding token through a free logit per (token, context-token) pair, i.e., for $\theta \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$,

$$(\text{tabular policy}) \quad \pi_{\theta}(\mathbf{y}_l | \mathbf{x}, \mathbf{y}_{<l}) = \pi_{\theta}(\mathbf{y}_l | \mathbf{y}_{l-1}) = \text{softmax}(\theta_{:, \mathbf{y}_{l-1}})_{\mathbf{y}_l}, \quad (5.3)$$

where $\theta_{:, \mathbf{y}_{l-1}} \in \mathbb{R}^{|\mathcal{V}|}$ denotes the column of θ indexed by the preceding token $\mathbf{y}_{l-1}$.

Optimization for RL. We analyze the tabular policy of Eq. 5.3 under the RL objective $\phi_{\text{RL}}(\theta)$ of Eq. 3.1, taking the small-step-size limit of policy gradient ascent so that the parameter trajectory obeys the gradient flow

$$\frac{d}{dt} \theta(t) = \nabla \phi_{\text{RL}}(\theta(t)), \quad t \geq 0 \quad (5.4)$$

where $\theta(t)$ collects the parameters of the policy $\pi_{\theta(t)}$ at training time t , with initial condition $\theta(0) = \theta_{\text{ref}}$.

Our measurements indicate that per-pattern success rates are approximately invariant over training. We attribute this to an asymmetry in optimization difficulty: the question-to-pattern map $\pi_{\theta}(\mathbf{r} | \mathbf{q})$ is far more readily adjusted than the pattern-to-answer map $\pi_{\theta}(\mathbf{a} | \mathbf{q}, \mathbf{r})$. We encode this observation as the following assumption.

Assumption 5.1. *The success rate for each reasoning pattern $\mathbf{r}_i \in \mathcal{R}$ to provide the correct answer $\mathbf{r}^*$, defined as $\pi_{\theta}(\mathbf{a}^* | \mathbf{q}, \mathbf{r}_i)$, remains constant during training.*

Throughout the remainder of this section we impose Assumption 5.1 and write $p^*(\mathbf{r}) = \pi_{\theta}(\mathbf{a}^* | \mathbf{q}, \mathbf{r})$ for the (time-invariant) success rate of pattern $\mathbf{r}$. We proceed in two steps: we first characterize the maximizer of the RL objective (Eq. 3.1) under the general autoregressive policy (Eq. 5.2), and then analyze the RLVR and RLIF gradient-flow dynamics under the tabular policy (Eq. 5.3).

5.2 The Optimal Policy for RL

The measurements of the previous section show that RLVR raises accuracy steadily by up-weighting higher-success-rate patterns whereas RLIF improves accuracy only transiently. As a first step toward explaining this contrast, we characterize the objective maximizer for both reward specifications.

Proposition 5.2. *Suppose we maximize the RL objective (Eq. 3.1) using a general autoregressive policy (Eq. 5.2) and Assumption 5.1 holds. Then, the optimal policy satisfies:*

$$\pi_{\text{opt}}(\mathbf{r} | \mathbf{q}) = \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r} | \mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}, \quad (5.5)$$

where $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r} | \mathbf{q})$ is the normalizing coefficient, and $R(\mathbf{r})$ denotes the reasoning path reward. Specifically:

- For RLVR, $R(\mathbf{r})$ equals the success rate of reasoning pattern $\mathbf{r}$, i.e., $R_{\text{RLVR}}(\mathbf{r}) = p^*(\mathbf{r})$.
- for RLIF, $R(\mathbf{r})$ equals to the confidence for the reasoning pattern $\mathbf{r}$ for final answer distribution, i.e., $R_{\text{RLIF}}(\mathbf{r}) = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a} | \mathbf{q}, \mathbf{r}))$.

Proposition 5.2 gives the maximizer in closed form for both objectives. Provided π_{opt} lies in the representable set of the general autoregressive policy—which holds for both practical LLMs and the tabular policies analyzed below—the RLVR maximizer up-weights pattern $\mathbf{r}$ relative to the reference exactly when its success rate is large enough that $\exp(\frac{1}{\beta} p^*(\mathbf{r}))/Z > 1$, so the reweighting is monotone in $p^*(\mathbf{r})$. For RLIF, by contrast, $R_{\text{RLIF}}(\mathbf{r})$ depends only on the confidence of the answer distribution and is blind to whether the mass concentrates on the correct answer $\mathbf{a}^*$ or on an incorrect alternative. Consequently the RLVR maximizer is guaranteed to be at least as accurate as the reference, whereas the RLIF maximizer carries no such guarantee.

Moreover, since the regularization coefficient is typically small in practice (e.g., $\beta = 0.001$) [26, 5, 44], taking the limit $\beta \rightarrow 0$ yields

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \lim_{\beta \rightarrow 0} \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) = \begin{cases} 1 & \mathbf{r} = \arg \max_{\mathbf{r}} R(\mathbf{r}) \\ 0 & \text{otherwise} \end{cases}, \quad (5.6)$$

so the maximizer degenerates to the point mass on the reward-maximizing pattern $\arg \max_{\mathbf{r}} R(\mathbf{r})$, independently of the reference policy’s pattern distribution. The characterization of the maximizer, however, is silent about the *path* taken to reach it: because the objective is non-convex, it explains neither how RLVR reliably locates high-reward patterns nor why RLIF can improve accuracy transiently before degrading. We resolve both questions in the next subsections by analyzing the tabular gradient-flow dynamics (Eq. 5.3) directly.

5.3 Training Dynamic Analysis for RLVR

Consistent with the measurement that RLVR’s gains scale with base-model quality, our analysis tracks the selection probability of the *optimal reasoning pattern*—the pattern $\mathbf{r}^* = \arg \max_{\mathbf{r}} p^*(\mathbf{r})$ of maximal success rate for reaching $\mathbf{a}^*$ —along the gradient flow. The dynamics partition into two regimes. In the first, $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$ increases monotonically and converges to 1. In the second, the trajectory first passes through an *entanglement stage* in which a suboptimal pattern $\mathbf{r}' \neq \mathbf{r}^*$ retards the growth of $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$; once this stage is cleared, the dynamics revert to those of the first regime and again converge to $\mathbf{r}^*$.

Theorem 5.3 (Regime 1: Sufficient Condition for Efficient Convergence). *Consider the RLVR (Eq 3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 holds. Let $\mathbf{r}^*$ be the optimal reasoning pattern, if the overall accuracy of the initialized model $\pi_{\theta_{\text{ref}}}$, defined as $\text{ACC}_{\theta_{\text{ref}}} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r})$, satisfies:*

$$(\text{Regime 1}) \quad \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*, \quad (5.7)$$

then for any $\epsilon > 0$, there exists $T_1 = \mathcal{O}(\frac{1}{\epsilon})$ such that for $t > T_1$, we have $1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) < \epsilon$.

Theorem 5.3 formalizes the high-quality regime: the reference model is strong enough that its overall accuracy dominates the success rate of every non-optimal pattern (Eq 5.7). Under this separation condition the gradient flow drives $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$ to within ϵ of 1 in time $\mathcal{O}(1/\epsilon)$, so the policy attains an accuracy approaching $p^*(\mathbf{r}^*)$.

Empirically, however, RLVR optimization can stall [44, 36], a failure mode associated with weaker reference models. This motivates the second regime, in which the trajectory traverses an entanglement stage where the *second-best pattern* $\mathbf{r}' = \arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$ competes with $\mathbf{r}^*$ and slows its growth.

Theorem 5.4 (Regime 2: Slow Convergence for optimal reasoning pattern). *Consider the RLVR (Eq 3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 holds. Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second optimal reasoning patterns, if the overall accuracy of the initialized model satisfies:*

$$(\text{Regime 2}) \quad p^*(\mathbf{r}') > \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}, \quad (5.8)$$

then there exists:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right), \text{ where } \gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|\mathbf{q})} \quad (5.9)$$

with constants C_1, C_2 depending on the success rates of reasoning patterns, such that for we can guarantee the model transform from regime 2 (Eq 5.8) to regime 1 (Eq 5.7), i.e., $\text{ACC}_{\theta(t)} > p^*(\mathbf{r}), \forall \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^* \text{ for } t \geq T_0$.

Theorem 5.4 isolates the tractable instance in which exactly two patterns— $\mathbf{r}^*$ and $\mathbf{r}'$ —have success rates exceeding the overall accuracy. We adopt this two-pattern separation to keep the analysis closed-form; it extends to the general setting in which *at least* two patterns exceed the average success rate, at the cost of a more involved argument.

The operative consequence of Theorem 5.4 is a time budget: RLVR needs T_0 steps to raise the overall accuracy above $p^*(\mathbf{r}')$ —equivalently, to enter the regime of Theorem 5.3 in which $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$ grows

efficiently. Whereas the convergence time T_1 of Theorem 5.3 is polynomial in $1/\epsilon$, the transition time T_0 can grow super-exponentially in $\gamma_{\pi_{\text{ref}}}$, the ratio of the aggregate success mass routed through $\mathcal{R}/\{\mathbf{r}'\}$ to that routed through $\mathbf{r}^*$. When the reference policy assigns $\mathbf{r}^*$ only negligible probability, $\gamma_{\pi_{\text{ref}}}$ becomes large and T_0 prohibitive, so the policy can spend an extremely long horizon before selecting $\mathbf{r}^*$ with appreciable probability—the behavior we term the entanglement stage.

Section 6 supplies case studies and simulations that instantiate both regimes.

5.4 Theoretical Explanation for RLIF

Proposition 5.2 showed that the RLIF reward, being a function of answer-distribution confidence alone, cannot discriminate the correct answer $\mathbf{a}^*$ from incorrect alternatives, so at convergence RLIF may degrade accuracy relative to the base model—matching the non-monotone trajectories observed empirically. Yet prior work reports that RLIF does improve performance without external rewards Agarwal et al. [4], Zhao et al. [47]. We reconcile the two by proving that RLIF increases accuracy at the *start* of training for a sufficiently well-trained base model.

We formalize “well-trained” through the following majority-correctness assumption.

Assumption 5.5. *For the base model, the correct answer $\mathbf{a}^*$ has the highest probability across all possible answers, i.e., $\arg \max_{\mathbf{a} \in \mathcal{A}} \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\text{base}}(\mathbf{a}|\mathbf{q}, \mathbf{r}) \pi_{\text{base}}(\mathbf{r}|\mathbf{q}) = \mathbf{a}^*$.*

Because majority voting [35] is empirically effective, a well-trained LLM places more marginal mass on $\mathbf{a}^*$ than on any incorrect answer, so Assumption 5.5 is expected to hold for modern LLMs. Under this assumption we specialize to a binary answer space $|\mathcal{A}| = 2$, success rates drawn i.i.d. from a uniform distribution, and a maximum-entropy pattern selector², and evaluate the accuracy derivative at initialization $t = 0$. This yields the following theorem.

Theorem 5.6 (RLIF Increases Overall Accuracy at Initialization). *Consider the RLIF (Eq 5.1) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 and 5.5 hold, $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = \frac{1}{|\mathcal{R}|}$ for all $\mathbf{r} \in \mathcal{R}$, $|\mathcal{A}| = 2$ and success rate for each reasoning path follows a uniform distribution $p^*(\mathbf{r}) \sim U[0, 1]$. Then when $|\mathcal{R}| \rightarrow +\infty$, the following holds:*

1. *The accuracy derivative at $t = 0$ is positive with probability 1, i.e. $P(\frac{d}{dt} \text{ACC}_{\theta(t)}|_{t=0} > 0) = 1$.*
2. *With probability $p = 0.5$, that $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$.*

The first claim of Theorem 5.6 establishes a strictly positive accuracy derivative at $t = 0$, which is the mechanism behind RLIF’s early gains. The long-horizon behavior is governed by Proposition 5.2: at $\beta = 0$ the policy converges to the point mass on $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r})$. The second claim shows that, with probability 0.5, this reward-maximizing pattern is precisely the accuracy-minimizing one, $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$. The two claims together resolve the apparent contradiction: RLIF raises accuracy initially (positive derivative) but, with probability one-half, converges to the least accurate pattern and thus to a fixed point below base-model accuracy.

6 Case Studies and Numerical Simulations

To validate and interpret the theory, we construct controlled case studies and Monte Carlo simulations that instantiate each theorem. For RLVR, we realize the two regimes of Theorems 5.3 and 5.4 directly in the gradient-flow dynamics, shown in Fig. 2 (upper). For RLIF, we run repeated simulations with per-pattern success rates sampled from a uniform distribution and measure both the initial-step accuracy change and the convergence fixed point, testing the two claims of Theorem 5.6; results appear in Fig. 2 (lower).

6.1 Case Studies for RLVR

Section 5.3 established two regimes for the RLVR dynamics. In the first (Eq 5.7), $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$ increases monotonically to convergence; in the second (Eq 5.8), the trajectory traverses an entanglement stage of duration T_0 before reverting to first-regime dynamics and converging to $\mathbf{r}^*$. We instantiate each condition:

²Prior work [33] shows that RL acts predominantly on high-entropy tokens; we take the extremal case $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = 1/|\mathcal{R}|$ for all $\mathbf{r} \in \mathcal{R}$.

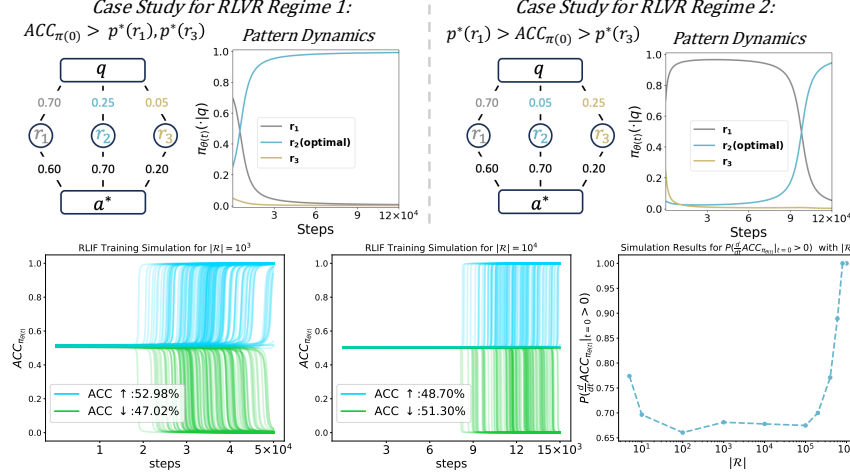


Figure 2: RLVR case studies (upper) and RLIF numerical simulations (lower). Upper: the two RLVR regimes of Section 5.3—efficient monotone convergence (left) and the entanglement-stage slowdown (right). Lower: the empirical probability that RLIF converges to an improved versus a degraded fixed point (left) and the probability of a positive accuracy change at the first training step (right), each as a function of the number of candidate patterns $|\mathcal{R}|$.

- Fig. 2 (upper left) sets the reference model to satisfy Eq 5.7, i.e., overall accuracy exceeds the success rate of every non-optimal pattern; the selection probability of r^* then rises monotonically and converges rapidly, as predicted by Theorem 5.3.
- Fig. 2 (upper right) instantiates Theorem 5.4: the second-best pattern r' has success rate above the initial overall accuracy, so the trajectory enters the entanglement stage and must clear the transition time T_0 before the rapid first-regime dynamics resume. The configuration sets $\gamma_{\pi_{\text{ref}}}$ (Eq 5.9) to the large value $\gamma_{\pi_{\text{ref}}} = 6$ (since $(0.25 + 0.05)/0.05 = 6$); because T_0 scales as $\gamma_{\pi_{\text{ref}}}^{\gamma_{\pi_{\text{ref}}}}$, convergence to r^* is severely delayed. The simulated pattern dynamics exhibit exactly this entanglement stage, in quantitative agreement with the theory.

6.2 Numerical Simulations For RLIF

Section 5.4 predicted that, under Assumption 5.5 and the base-model distribution of Theorem 5.6, the accuracy derivative at $t = 0$ is positive with probability approaching 1 as $|\mathcal{R}| \rightarrow \infty$, while the convergence fixed point is the least accurate pattern with probability $1/2$. The simulations confirm both predictions. In Fig. 2 (lower left), sweeping $|\mathcal{R}|$, the empirical probability of converging to an improved fixed point ($ACC \uparrow$) matches that of converging to a degraded one ($ACC \downarrow$), corroborating the second claim. In Fig. 2 (lower right), the measured probability of a positive first-step accuracy change increases toward 1 with $|\mathcal{R}|$, corroborating the first claim.

7 Conclusions and Limitations

This work characterizes the RL training dynamics of LLMs by coupling controlled empirical measurement with a formal dynamical analysis. Working within a two-stage $q \rightarrow r \rightarrow a$ factorization of generation, we derive the gradient-flow behavior of two canonical reward specifications, RLVR and RLIF, and confirm each prediction through case studies and Monte Carlo simulations. Three limitations delimit the scope of the analysis and set the agenda for future work: the reasoning patterns are induced by an LLM classifier and their semantic validity requires independent verification; the two-stage framework abstracts away structure present in real reasoning traces and should be extended to richer generative processes; and the convergence results are stated under specific base-model conditions that could be relaxed toward more general assumptions.

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A Additional Experiments

A.1 Pattern Distribution Dynamics on RL-enhanced Qwen2.5-3B

Section 4.2 traced the pattern-distribution dynamics on Number Theory problems from MATH. Here we replicate the same protocol on Geometry problems (Figure 3). The trajectories reproduce the main-paper behavior: under RLVR the policy concentrates probability mass on the higher-accuracy patterns as training proceeds, whereas under RLIF the pattern frequencies remain unstable—consistent with both the empirical and the theoretical results in the main text.

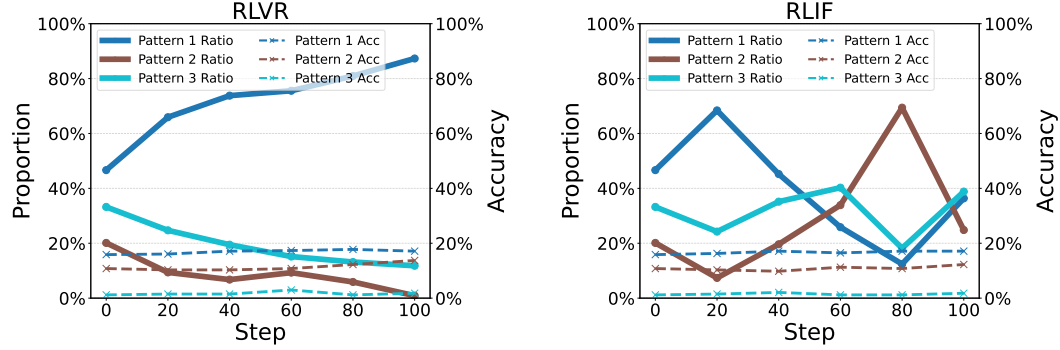


Figure 3: Joint dynamics of pattern frequency and per-pattern accuracy for RL-trained models on Geometry problems. Under RLVR the policy concentrates on the highest-accuracy pattern (Pattern 1); under RLIF the frequencies fail to stabilize.

A.2 Qwen-2.5-7B-Instruct vs. Qwen-2.5-7B-SimpleRL-Zoo

To test whether the findings transfer across model families, we replicate the analysis on an independent model pair (Figure 4). We take Qwen-2.5-7B-Instruct [39] as the base model and Qwen-2.5-7B-SimpleRL-Zoo [44]—an open-source variant RLVR-trained for mathematical reasoning—as its RL-enhanced counterpart. The evaluation covers five MATH [16] domains (number theory, geometry, algebra, calculus, and counting and probability) using high-difficulty problems, and additionally a harder problem drawn from AMC23 [1]. The measurements are consistent with the main results: RLVR raises the frequency of the high-accuracy patterns and suppresses the less effective ones. Per-task breakdowns are given in Appendix B.3.

Across all tasks, the highest-success-rate patterns become uniformly more prevalent after RLVR, reinforcing the conclusions of Section 4 and Section 5. Notably, on the Algebra and AMC23-19 tasks of Figure 4 the per-pattern success rates are essentially unchanged between the two models, which not only matches our earlier observation but also provides direct empirical support for the time-invariance posited in Assumption 5.1.

A.3 Qwen-2.5-32B-Instruct vs. QwQ-32B

To probe scalability, we repeat the analysis on a larger model, QwQ-32B [32], which is derived from Qwen-2.5-32B-Instruct [39] and substantially strengthened for reasoning via RLVR. We measure the pattern-distribution shift between the two models on four tasks: number theory and geometry problems from MATH [16], a problem from AIME 2024 [7], and a problem from OlympiadBench [15].

The RLVR-trained QwQ-32B concentrates on the patterns that are most accurate under Qwen-2.5-32B, in agreement with the theory. We omit per-pattern accuracy for QwQ because these models are queried through an API that does not expose the chain-of-thought needed to continue generation from a fixed prefix; without it, the pattern-conditional success rate (as opposed to the model’s end-to-end accuracy) is not identifiable. See “Accuracy Analysis” in Appendix B.2 for the procedure this requires.

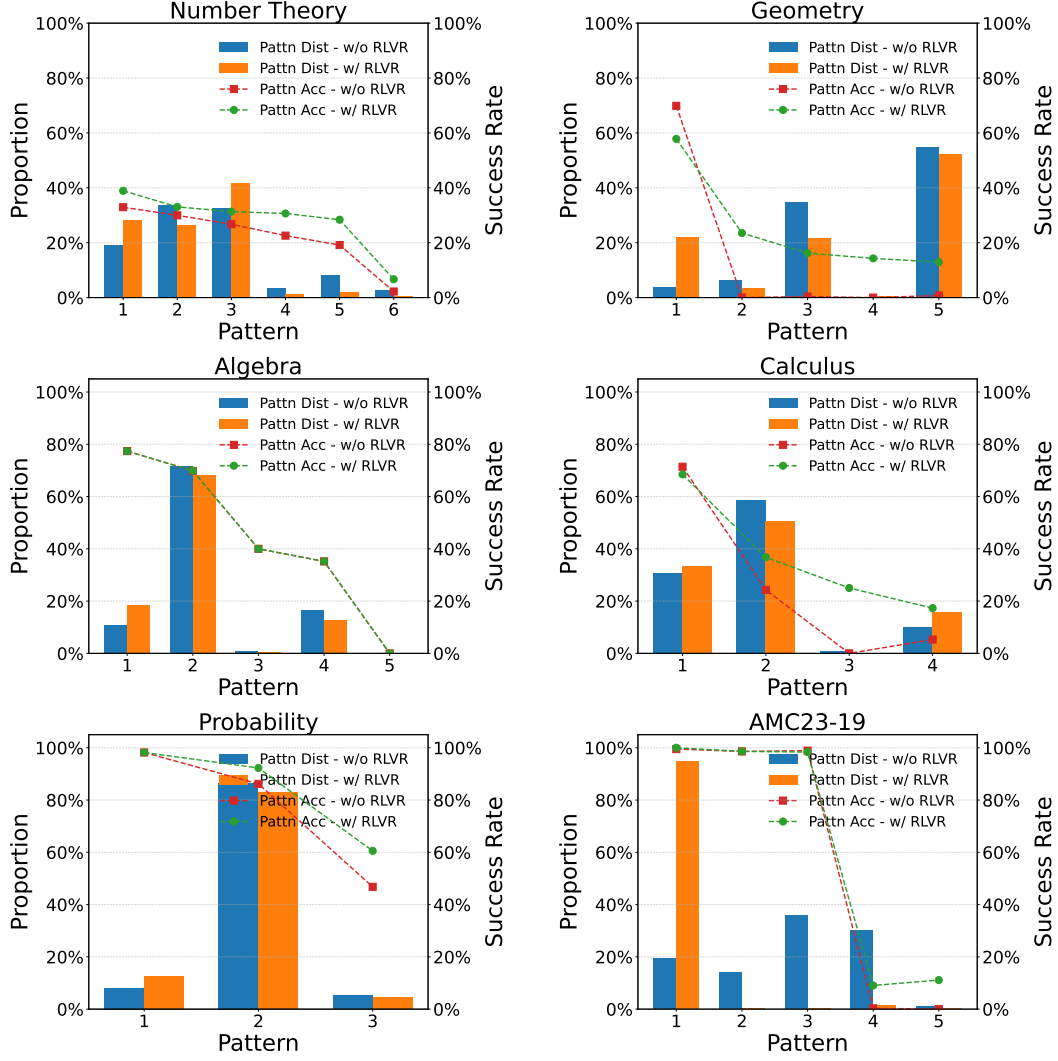
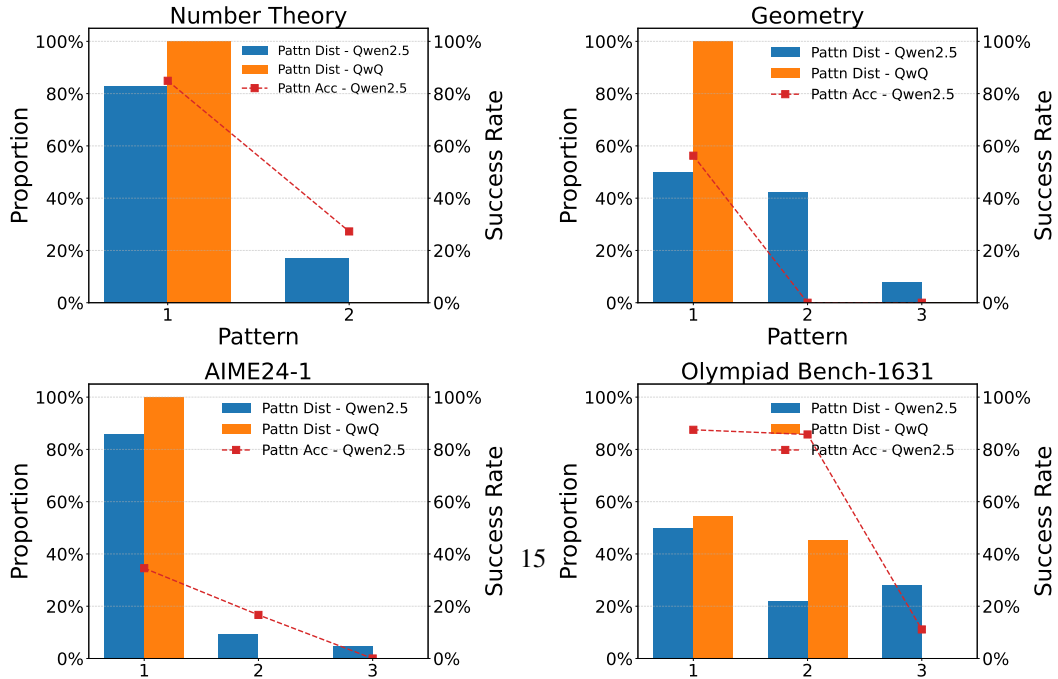


Figure 4: Pattern frequencies and corresponding success rates for the base and RLVR-enhanced models across the additional tasks, in agreement with the empirical and theoretical results. Bars (Pattern Dist) report each pattern’s relative frequency among all patterns; dots (Pattern Acc) report its success rate.



546 B Experimental Details

547 B.1 Experimental Setup

548 **Entire Training Procedures:** We fix the base model to Qwen2.5-3B and the training set to MATH,
549 and run both RLVR and RLIF. Optimization uses the verl [31] framework with the hyperparameter
550 configuration of the framework’s MATH example held fixed across both methods; RLVR and RLIF
551 differ only in the reward computation. For each method we obtain three independent training curves
552 by varying the random seed.

553 **Token-level Analysis:** This experiment uses Qwen2.5-3B and its RL-enhanced counterparts above.
554 We draw 400 questions uniformly at random from each of the GSM8K and MATH test sets and
555 use the full AIME24 set. For each question we first sample a base-model answer, then form a new
556 prompt by concatenating the question with that answer and feed it to the RL-enhanced model. Via the
557 OpenAI library interface we read off, for every token in the prompt, its conditional probability and
558 its rank among all vocabulary entries at that position. The reported statistic is the fraction of tokens
559 whose rank differs between the base and RL-enhanced models.

560 **Reasoning-pattern Level Analysis:** We run two variants. The first measures the pattern-distribution
561 dynamics *during* training, using the Qwen2.5-3B checkpoints and their RL-enhanced versions
562 evaluated at 20-step intervals. The second tests transfer across base models and scale on the open-
563 source pairs Qwen-2.5-7B-SimpleRL-Zoo and QwQ-32B together with their base models; because
564 only pre- and post-RL checkpoints are available for these pairs, this variant reports the before/after
565 change in the pattern distribution rather than its full training trajectory.

566 B.2 Reasoning Patterns Analysis Pipeline

567 **Model Inference:** We sample questions across task types (e.g., Geometry, Algebra) and difficulty
568 levels (Level 4, Level 5) from MATH and related sources. Per question we draw 1024 responses for
569 the pattern-dynamics experiment, and between 64 and 1024 responses (depending on task category)
570 for the before/after experiment. These samples constitute the corpus for the subsequent extraction
571 and classification stages.

572 **Reasoning Patterns Extraction:** From a subset of the base-model responses we prompt the GPT-4o
573 [19] API to induce the pattern taxonomy; the exact prompt is reproduced in Appendix D.1. To make
574 the induced taxonomy deterministic we set the GPT-4o decoding temperature to 0.

575 **Responses Classification:** We then assign every response to one of the induced categories with
576 GPT-4o, again at temperature 0; the classification prompt is given in Appendix D.2.

577 **Accuracy Analysis:** To isolate the effect of the pattern distribution on accuracy from any change in
578 the reasoning process itself, we hold the reasoning trace fixed and vary only the pattern. Concretely,
579 we take a base-model response, delete the sentence containing the final answer, append the remaining
580 prefix to the question, and present the result as a new prompt to the RL-enhanced model, which then
581 completes the answer. Since the prefix commits the response to a specific reasoning pattern, the
582 completion’s correctness serves as an estimate of that pattern’s success rate. Any resulting accuracy
583 change is thus attributable to the pattern conditioning rather than to a change in the reasoning trace.

584 B.3 Details for Each Task

585 Below we specify, for each task, the problem definition and representative reasoning patterns (Table 2);
586 the corresponding prompt templates are given in Appendix D.3.

587 **Number Theory:** Each instance specifies a set of bags initially holding equal coin counts; after
588 additional coins are added and redistributed equally, the total must exceed a stated threshold while
589 preserving equal distribution, and the model must return the minimum initial per-bag count. We
590 instantiate the task from a level-5 MATH problem (id: test/number_theory/1055) [16], generating
591 a 32-instance evaluation set by varying parameters such as bag count and coin total. Prompts use

Task	Reasoning Patterns Examples	Common Elements / Key Words
Number Theory	Modular Congruence with Coefficient Simplification	Uses modular arithmetic to express divisibility conditions.
	Inequality-Driven Search for Minimal Solution	Often involves substituting back to compute the total coins.
Geometry	Systematic Inequality Application	Explicitly lists and solves each of the three inequalities.
	Verification of Scalene Condition	Adding an extra layer of validation to verifies the third side length.
Algebra	Iterative Floor Division with Leftover Tracking.	Uses floor division to compute new cans per step.
	Recursive Recycling with Aggregated Leftovers.	Combines leftovers with newly produced cans before recycling.
Calculus	Direct Simplification and Principal Value Matching	Assumes inputs fall within the principal range.
	Interval Analysis with Case Splitting	Splits the whole domain into different intervals.
Counting and Probability	Direct Probability Setup and Quadratic Solution	Explicitly calculates combinations for total and favorable outcomes.
	Early Simplification and Cross-Multiplication	Early cross-multiplication to eliminate denominators.
AMC23-Q19	Prime Factorization and Simplification	Counting Digits in Large Numbers.
	Rewriting (8^5) as (2^{15}) (15^5) as $(3^5 \cdot 5^5)$.	Final step of counting digits: $(3 \text{ (from 243)} + 15 \text{ (zeros)}) = 18$.
AIME24-Q1	Direct Equation Setup and Elimination	Straightforward and relies on algebraic manipulation.
	Alternative Equation Formulation and Solving	Expressing variables in terms of others early on.
OlympiadBench-Q1631	Direct Calculation and Empirical Testing	Relying on direct computation and empirical verification.
	Factorization-Based Reasoning	Using algebraic factorization to argue certain terms y_n .

Table 2: Representative reasoning patterns and their characteristic elements for each evaluation task.

one-shot formatting with a simple, unrelated exemplar that fixes the “boxed{ }” answer format without biasing the reasoning strategy. For pattern extraction we select 20 questions and 4 responses each; for the distribution comparison we sample 64 responses per question and classify all of them under the extracted taxonomy, yielding the before/after pattern distributions.

Geometry: Each instance supplies two side lengths of a scalene triangle and asks for the number of admissible integer (centimeter) lengths of the third side. The task is instantiated from a level-4 MATH geometry problem (id: test/geometry/1046). The evaluation pipeline mirrors Number Theory: a 32-instance synthetic set, 20 questions with 4 base-model (Qwen2.5-7b-Instruct) responses each for pattern extraction, and a before/after comparison over the full set at 64 responses per question.

Algebra: The algebra task is a can-recycling problem: given an initial stock of used cans and a recycling ratio in which n used cans yield one new can, the model must compute the total number of cans producible. Its difficulty lies in the iterative accounting: at each recycling step the remainder that is not evenly divisible carries over and may combine with later cans for further recycling. The task is instantiated from a level-4 MATH algebra problem (id: test/algebra/2768). All supplementary tasks from this point on inherit the settings of the preceding tasks (dataset size, sample counts, etc.).

Calculus: The calculus task specifies a single-variable trigonometric equation together with the variable’s domain and asks for the number of solutions on that domain. Though elementary, it probes the model’s basic pre-calculus competence. It is instantiated from a level-4 MATH pre-calculus problem (id: test/precalculus/1140).

611 **Counting and Probability:** The counting-and-probability task considers an urn of white and
612 black balls from which two balls are drawn at random. Given the probability that the draw yields
613 one ball of each color and the count of one color, the model must compute the minimum count
614 of the other color. It is instantiated from a level-4 MATH counting-and-probability problem (id:
615 test/counting_and_probability/79).

616 **AMC23 Question:** We use question 19 of AMC23 [1], drawing 48 responses for pattern extraction
617 and 1024 responses for the pattern-distribution analysis.

618 B.4 Details for Experiments on QwQ-32B and Qwen-2.5-32B

619 We evaluate four tasks. Because these models are queried through an API, we fix a single question
620 per task, sample 64 responses from each of the two models, and run pattern extraction and classifi-
621 cation directly on those responses. The questions are: problem 1055 (number theory) and problem
622 1046 (geometry) from MATH, problem 1 from AIME24, and problem 1631 from OlympiadBench.
623 Representative patterns are listed in Table 2.

624 C Deferred Proofs

625 This appendix collects the proofs of the main theoretical results, presented in order: Proposition 5.2,
626 Theorem 5.3, Theorem 5.4, and Theorem 5.6.

627 C.1 Proof of Proposition 5.2

628 *Proof of Proposition 5.2.* The argument adapts the derivation of [26]. Recall the KL-regularized RL
629 objective

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_{\theta}(\mathbf{y} | \mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.1})$$

630 Under the two-stage framework of Section 5.1, the policy first samples $\mathbf{r}_i \in \mathcal{R}$ from $\pi_{\theta}(\mathbf{r}_i | \mathbf{q})$ and
631 then emits $\mathbf{a} \in \mathcal{A}$ from $\pi_{\theta}(\mathbf{a} | \mathbf{r}_i)$; marginalizing over the answer stage lets us write the objective as

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\mathbf{y} | \mathbf{q})} \left[\sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q}) \right] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.2})$$

632 Let $R(\mathbf{r}) = \sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q})$, then

$$\begin{aligned} \phi_{\text{RL}}(\theta) &= \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot | \mathbf{q}), \mathbf{a} \sim \pi_{\theta}(\cdot | \mathbf{r})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\theta}(\mathbf{r} | \mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a} | \mathbf{r})}{\pi_{\text{ref}}(\mathbf{r} | \mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a} | \mathbf{r})} \right) \right] \\ &= \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot | \mathbf{q})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\theta}(\mathbf{r} | \mathbf{q})}{\pi_{\text{ref}}(\mathbf{r} | \mathbf{q})} \right) \right] \\ &= -\beta \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot | \mathbf{q})} \left[\ln \left(\frac{\pi_{\theta}(\mathbf{r} | \mathbf{q})}{\frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\text{ref}}(\mathbf{r} | \mathbf{q})} \right) - \ln Z \right] \\ &= -\beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r} | \mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] + \beta \ln Z. \end{aligned}$$

633 where the third equality invokes Assumption 5.1 and $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp \left(\frac{1}{\beta} p^*(\mathbf{r}) \right) \pi_{\theta_{\text{ref}}}(\mathbf{r} | \mathbf{q})$ is the
634 partition function chosen so that

$$\pi^*(\mathbf{r} | \mathbf{q}) = \frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\theta}(\mathbf{r} | \mathbf{q}),$$

635 is a normalized distribution, i.e. $\sum_{\mathbf{r} \in \mathcal{R}} \pi^*(\mathbf{r} | \mathbf{q}) = 1$; note Z does not depend on $\mathbf{r}$.

636 Because Z is independent of π , maximizing the objective in Equation C.2 is equivalent to

$$\begin{aligned} &\max_{\pi} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi(\mathbf{y} | \mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] \\ &= \min_{\pi} \beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r} | \mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] - \beta \ln Z. \end{aligned}$$

637 Since the KL divergence is minimized precisely when its two arguments coincide, the maximizer of
 638 the KL-regularized objective is

$$\pi_{opt}(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}.$$

639 Specializing the reward term $R(\mathbf{r})$: for RLVR we have $R_{\text{RLVR}}(\mathbf{r}) = p^*$, while for RLIF

$$\begin{aligned} R'(\mathbf{r}) &= - \left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_{\theta}(\mathbf{r}_i|\mathbf{q})) + \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(|\mathcal{A}| \cdot \pi_{\theta}(\mathbf{a}_j|\mathbf{q}, \mathbf{r})) \right) \\ &= - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j|\mathbf{q}, \mathbf{r})) - \underbrace{\left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_{\theta}(\mathbf{r}_i|\mathbf{q})) + \log|\mathcal{A}| \right)}_c, \end{aligned}$$

640 where the bracketed term c is constant across $\mathbf{r} \in \mathcal{R}$. Substituting into the maximizer, the RLIF-
 641 optimal policy is

$$\begin{aligned} \pi_{opt}(\mathbf{r}|\mathbf{q}) &= \frac{1}{Z} \exp\left(\frac{1}{\beta} R'(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \\ &= \frac{\exp\left(\frac{1}{\beta} R'(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp\left(\frac{1}{\beta} (R'(\mathbf{r}_i))\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}_i|\mathbf{q})} \\ &= \frac{\exp\left(\frac{1}{\beta} (R'(\mathbf{r}) - c)\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp\left(\frac{1}{\beta} (R'(\mathbf{r}_i) - c)\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}_i|\mathbf{q})} \end{aligned}$$

642 since the constant c cancels between numerator and denominator. Hence $R_{\text{RLIF}}(\mathbf{r}) = R'(\mathbf{r}) - c =$
 643 $-\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j|\mathbf{q}, \mathbf{r}))$, which completes the proof. $\square$

644 C.2 Dynamics of Tabular Policy

645 Under the RL objective (Eq 3.1), the tabular policy (Eq 5.3), and Assumption 5.1, the gradient flow
 646 of θ evaluates to

$$\begin{aligned} \frac{d}{dt} \theta(t) &= \nabla \phi_{\text{RL}}(\theta(t)) \\ &= \nabla \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q})] \\ &= \sum_{\mathbf{r} \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q}) \nabla \pi_{\theta(t)}(\mathbf{r}|\mathbf{q})], \end{aligned}$$

647 where $\pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) = \text{softmax}(\theta_{:,q})_{\mathbf{r}}$, $\theta_{:,q} \in \mathbb{R}^{|\mathcal{V}|}$ is the column of θ indexed by $\mathbf{q}$, and $r_{\phi}(\mathbf{r}|\mathbf{q})^{KL}$
 648 denotes the KL-regularized reward. Differentiating the softmax gives

$$\begin{aligned} \frac{\partial \theta_{\mathbf{r}_i, \mathbf{q}}(t)}{\partial t} &= \sum_{\mathbf{r}_j \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) (\nabla \text{softmax}(\theta_{:,q})_{\mathbf{r}_j})_{\mathbf{r}_i}] \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) (-\pi_{\theta(t)}(\mathbf{r}_j|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}))] \\ &\quad + r_{\phi}^{KL}(\mathbf{r}_i|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \end{aligned}$$

649 where the overall accuracy is $\text{ACC}_{\theta(t)} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r})$. By the chain rule, the induced
 650 dynamics of the selection probability $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$ are

$$\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) = \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial \theta_{\mathbf{r}_j, \mathbf{q}}(t)} \cdot \frac{\partial \theta_{\mathbf{r}_j, \mathbf{q}}(t)}{\partial t}$$

651 C.3 Proof of Theorem 5.3

652 The proof of Theorem 5.3 proceeds in three steps: we show that the accuracy is non-decreasing,
 653 $\text{ACC}_{\theta(t)} \geq \text{ACC}_{\theta(0)}$ for $t \geq 0$ (Eq C.4); we then lower-bound $\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})$ (Eq C.5); and finally we
 654 convert this bound into the convergence-time estimate for t asserted in Theorem 5.3.

655 *Proof of Theorem 5.3.* At $\beta = 0$, Eq C.3 reduces to

$$\frac{\partial \theta_{\mathbf{r}_i, \mathbf{q}}(t)}{\partial t} = \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \cdot (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}). \quad (\text{C.3})$$

656 The selection-probability dynamics then follow as

$$\begin{aligned} \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial \theta_{\mathbf{r}_j, \mathbf{q}}(t)} \cdot \frac{\partial \theta_{\mathbf{r}_j, \mathbf{q}}(t)}{\partial t} \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}), \end{aligned}$$

657 and, differentiating the accuracy along the flow,

$$\begin{aligned} \frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &= \sum_{\mathbf{r}_i, \mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} \text{ACC}_{\theta(t)} \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \\ &\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) p^*(\mathbf{r}_i) \\ &= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)})^2 \geq 0. \end{aligned} \quad (\text{C.4})$$

658 Since $\frac{d}{dt} \text{ACC}_{\theta(t)} \geq 0$, the accuracy is non-decreasing and $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$ for $t \geq 0$. Writing
 659 $\mathbf{r}'$ for the second-best pattern $\mathbf{r}' = \arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$, the growth of $\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})$ is bounded below
 660 by

$$\begin{aligned} \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) \\ &\geq (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}')) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) \\ &= (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\ &\geq (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 \geq 0. \end{aligned} \quad (\text{C.5})$$

661 Since $\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \geq 0$, we have $\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \geq \pi_{\theta(0)}(\mathbf{r}^*|\mathbf{q})$ for $t \geq 0$. Setting $C =$
 662 $(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(0)}^2(\mathbf{r}^*|\mathbf{q})$ yields the autonomous lower bound

$$\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \geq C (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2.$$

663 Integrating this differential inequality gives

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) > 1 - \frac{1}{Ct + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^*|\mathbf{q})}}.$$

Consequently, for any $\epsilon > 0$, taking $T_1 = \frac{1}{\epsilon} \left(\frac{1}{\epsilon} - \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^*|\mathbf{q})} \right) = \mathcal{O}(\frac{1}{\epsilon})$ guarantees

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) > 1 - \epsilon.$$

for all $t > T_1$, which completes the proof. $\square$

C.4 Proof of Theorem 5.4

The proof of Theorem 5.4 has three steps: an upper bound on $\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})$ (Eq C.6); an analysis of the ratio $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}$ (Eq C.7); and a demonstration that $\text{ACC}_{\theta(t)} \geq p^*(\mathbf{r}')$ once $t > T_0$.

Proof of Theorem 5.4. Let $\mathbf{r}^*$ and $\mathbf{r}'$ denote the optimal and second-best patterns. The growth of $\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})$ is controlled by

$$\begin{aligned} \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\ &\leq \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}') - p^*(\mathbf{r}')) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\ &\leq \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 + p^*(\mathbf{r}') \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 \\ &\leq 2 (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2. \end{aligned}$$

Integrating this bound yields

$$\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \leq 1 - \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}. \quad (\text{C.6})$$

Now define the ratio $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}$. For every $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}$ its derivative satisfies

$$\begin{aligned} \frac{d}{dt} \rho_{\theta(t)}(\mathbf{r}_i) &= \frac{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}{\pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q})} \\ &= \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})} ((p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\ &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \leq 0 \end{aligned}$$

where the final inequality uses the Regime-2 condition together with $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$, which give $p^*(\mathbf{r}_i) < \text{ACC}_{\theta(t)}$ for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*\}$ and $t \geq 0$. Since $\frac{d}{dt} \rho_{\theta(t)}(\mathbf{r}_i) \leq 0$, the ratio is non-increasing, and therefore

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) = \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i)} \geq \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i)}$$

On the sub-interval where $\text{ACC}_{\theta(t)} < p^*(\mathbf{r}')$, writing $\gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|\mathbf{q})}$, we obtain

$$\begin{aligned} \frac{d}{dt} \rho_{\theta(t)}(\mathbf{r}_i) &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\ &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\gamma_{\pi_{\text{ref}}}} \\ &\leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}} \end{aligned}$$

677 that is,

$$\frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) \leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}, \quad (\text{C.7})$$

678 Integrating this differential inequality gives

$$\rho_{\theta(t)}(\mathbf{r}_i) \leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2t)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \rho_{\theta(0)}(\mathbf{r}_i).$$

679 where $\Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}')$. The accuracy gap over $p^*(\mathbf{r}')$ can be decomposed as

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &= \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r}) - p^*(\mathbf{r}') \\ &= p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) + \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}_i) - p^*(\mathbf{r}')) \right] \\ &\geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right]. \end{aligned}$$

680 Setting $C_2 = 1/\Delta$ and $C_1 = p^*(\mathbf{r}')/\Delta$, and defining

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right),$$

681 the ratio-sum at $t = T_0$ is bounded by

$$\begin{aligned} &\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(T_0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\ &\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2T_0)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\ &\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{\left(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) \cdot \left(\frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) \\ &= \frac{1}{\left(\left(\frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \gamma_{\pi_{\text{ref}}} \\ &= \Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}'). \end{aligned}$$

682 Substituting this into the accuracy decomposition, for $t > T_0$ we conclude

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &\geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right] \\ &\geq p^*(\mathbf{r}^*) [(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - (p^*(\mathbf{r}^*) - p^*(\mathbf{r}'))] = 0. \end{aligned}$$

683 This concludes the proof of the theorem. $\square$

684 C.5 Proof of Theorem 5.6

685 *Proof of Theorem 5.6.* In the specialized setting $|\mathcal{A}| = 2$ and $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = \frac{1}{|\mathcal{R}|}$, Assumption 5.5 is
 686 equivalent to $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r}) > 0.5$. The accuracy derivative expands as

$$\begin{aligned} \frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \\ &= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \left(- \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a} | \mathbf{r}_i, \mathbf{q})) - \text{Con}_{\theta(t)} \right), \end{aligned}$$

687 where $\text{Con}_{\theta(t)} = - \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a} | \mathbf{r}_i, \mathbf{q}))$. Writing $\mathcal{A} = \{\mathbf{a}^*, \mathbf{a}'\}$ with $\mathbf{a}'$
688 the incorrect answer and evaluating at the uniform initialization $\pi_{\theta(0)}(\mathbf{r}_i | \mathbf{q}) = \frac{1}{|\mathcal{R}|}$, the derivative at
689 $t = 0$ becomes

$$\begin{aligned} \left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} &= \sum_{\mathbf{r}_i \in \mathcal{R}} \frac{1}{|\mathcal{R}|^2} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \\ &\quad \left(- \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_i, \mathbf{q})) + \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_j, \mathbf{q})) \right) \\ &= \frac{1}{|\mathcal{R}|^2} \sum_{\mathbf{r}_i \in \mathcal{R}} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \left(- \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_i, \mathbf{q})) \right). \end{aligned}$$

690 Consider the limit $|\mathcal{R}| \rightarrow +\infty$ with success rates $p^*(\mathbf{r})$ drawn i.i.d. from $U[0, 1]$. By the central limit
691 theorem the sample mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$ concentrates around its expectation 0.5 with fluctuations
692 of order $\mathcal{O}(1/\sqrt{|\mathcal{R}|})$; denote this deviation by $\delta = \mathcal{O}(1/|\mathcal{R}|^{1/2}) > 0$, so that $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r}) =$
693 $0.5 + \delta$.

694 By the Gibbs conditioning principle ([12], Corollary 7.3.5 and Theorem 7.3.8), as $|\mathcal{R}| \rightarrow \infty$ the
695 empirical distribution of the $p^*(\mathbf{r})$ conditioned on the mean constraint $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r}) = 0.5 + \delta$
696 converges to the tilted density $f(x) = 1 + 12\delta(x - 0.5)$, $x \in [0, 1]$. Replacing the sum over $\mathbf{r}$ by the
697 corresponding integral against f , the derivative reduces to

$$\left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} = \int_0^1 - (1 + 12\delta(x - 0.5)) \cdot (x - 0.5 - \delta) \cdot \log(x(1 - x)) dx = \frac{2}{3} \delta > 0$$

698 Since $\delta > 0$, the derivative is strictly positive, and this holds with probability approaching 1 as
699 $|\mathcal{R}| \rightarrow \infty$; this establishes the first statement.

For the second statement, we first reduce the RLIF reward to a function of the success rate. With
 $|\mathcal{A}| = 2$, the reward of pattern $\mathbf{r}$ is the negative token-averaged log-likelihood

$$R_{\text{RLIF}}(\mathbf{r}) - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j | \mathbf{q}, \mathbf{r})) = -\frac{1}{2} \log(p^*(\mathbf{r})(1 - p^*(\mathbf{r})))$$

700 Because $g(p) = p(1 - p)$ attains its maximum at $p = 0.5$, maximizing $R_{\text{RLIF}}(\mathbf{r})$ is equivalent to
701 selecting the pattern whose success rate is farthest from 0.5.

Let $p_{\min}^* = \min_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$ and $p_{\max}^* = \max_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$. The reward-maximizing pattern is the one
attaining $p_{\min}^*$ exactly when $p_{\min}^*$ is farther from 0.5 than $p_{\max}^*$, i.e.

$$0.5 - p_{\min}^* > p_{\max}^* - 0.5 \iff p_{\min}^* + p_{\max}^* < 1$$

702 Hence the event $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$ coincides with the event $p_{\min}^* + p_{\max}^* < 1$.

703 It remains to compute the probability of this event. With success rates drawn from the limiting
704 density $f(x) = 1 + 12\delta(x - 0.5)$, order-statistics theory gives that the scaled extremes of $n = |\mathcal{R}|$
705 samples converge in distribution to independent exponential variables. Setting $Y_n = n \cdot p_{\min}^*$ and
706 $Z_n = n \cdot (1 - p_{\max}^*)$, as $n \rightarrow \infty$:

707 • Y_n converges to $Y \sim \text{Exp}(\lambda_Y)$, where the rate is the density at the lower bound: $\lambda_Y = f(0) =$
708 $1 - 6\delta$.

709 • Z_n converges to $Z \sim \text{Exp}(\lambda_Z)$, where the rate is the density at the upper bound: $\lambda_Z = f(1) =$
710 $1 + 6\delta$.

In terms of these asymptotic variables the condition $p_{\min}^* + p_{\max}^* < 1$ rewrites as

$$\frac{Y_n}{n} + \left(1 - \frac{Z_n}{n}\right) < 1 \implies Y_n < Z_n$$

so the event has probability $P(Y < Z)$. For two independent exponentials this equals

$$P(Y < Z) = \frac{\lambda_Y}{\lambda_Y + \lambda_Z} = \frac{1 - 6\delta}{(1 - 6\delta) + (1 + 6\delta)} = \frac{1 - 6\delta}{2} = \frac{1}{2} - 3\delta$$

Since $\delta = O(1/|\mathcal{R}|^{1/2}) \rightarrow 0$ as $|\mathcal{R}| \rightarrow \infty$, the limiting probability is

$$\lim_{|\mathcal{R}| \rightarrow \infty} P(\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})) = \lim_{\delta \rightarrow 0} \left(\frac{1}{2} - 3\delta\right) = \frac{1}{2}$$

711 which establishes the second statement.

712

□

713 D Full Prompts

714 D.1 Full Prompt for Reasoning Patterns Extraction

Full Prompt for Reasoning Patterns Extraction

You will analyze multiple AI reasoning processes, showing how different models solve problems.

Analysis Steps:

For each reasoning process, identify:

Key words and recurring phrases

Logical structure of the argument

Problem-solving techniques used

Step-by-step progression

Group similar reasoning processes into exactly 5 patterns based on shared:

Vocabulary patterns (common terms and phrases)

Logical frameworks (how arguments are structured)

Solution approaches (methods used to reach conclusions)

IMPORTANT! You need to categorize general thinking patterns for problem solving, rather than categorizing types of problems themselves. In other words, each type of thinking pattern you give can be applied to any problem, rather than being task-specific. What you need to classify is the way of thinking, not the way of solving problems.

For example, you can classify them into:

1. derivation based on existing knowledge

2. derivation combined with verification

3. independent calculation, suitable for learners starting from the basics,

4. detailed manual derivation, suitable for learners who need to deeply understand each step

5. etc

Or you can classify them into:

1. knowledge call type

2. derivation and verification type

3. etc.

Output Format:

Pattern 1:

Name and description

Defining characteristics

Example processes from input

Common elements

Pattern 2:

715

Name and description
 Defining characteristics
 Example processes from input
 Common elements
 ...
 There are reasoning processes:
 {extracted_responses}
 Group similar reasoning processes into exactly 5 patterns. Please provide a classification of reasoning patterns, with each group defined by their common characteristics and examples from these processes.

716

717 D.2 Full Prompt for Responses Classification

Full Prompt for Responses Classification

Please classify each reasoning process into one of the following five patterns. Output the classifications in JSON format.
 {patterns_categories}
 Here are the reasoning processes to classify:
 {extracted_responses}
 Please output your classification in this JSON format:

718

719 D.3 Prompt Examples for Evaluation Tasks

Prompt Example for Easy Countdown

<lim_start>system
 You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_end>
 <lim_start>
 user Using the numbers [1,2,3,7], create an equation that equals 5. You can only use basic arithmetic addition (+) and subtraction (-), and each number from the list can be used at most once. Show your work in <think> </think> tags. And return the final answer in <answer> </answer> tags, for example, to get 5 from [1,2,3,7], we have <answer> 2 + 3 = 5 </answer>; to get 13 from [1,2,3,7], we have <answer> 1 + 2 + 3 + 7 = 13 </answer>.<lim_end>
 <lim_start>assistant
 Let me solve this step by step.
 <think>

720

Prompt Example for Long Arithmetic

<lim_start>system
 You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_end>
 <lim_start>user
 Considering the experssion (((((((((16 + 47) + 92) + 84) + 5) + 49) + 82) - 33) - 99) - 69), you need to start from the innermost bracket and work your way out. You need to give the specific calculation process. You have to do a certain amount of calculations on each line, and print each calculation in the format of a + b = c or a - b = c, for example "1 + 2 = 3".<lim_end>
 <lim_start>assistant
 Let me solve this step by step.
 <think>

721

Prompt Example for Number Theory

<lim_start>system

You are a helpful assistant.<lim_end>

<lim_start>user

You have four bags of copper coins. Each bag has the same number of copper coins. One day, you find a bag of 23 coins. You decide to redistribute the number of coins you have so that all five bags you hold have the same number of coins. You successfully manage to redistribute all the coins, and you also note that you have more than 120 coins. What is the smallest number of coins you could have had before finding the bag of 23 coins?

Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>

<lim_start>assistant

722

Prompt Example for Geometry

<lim_start>system

You are a helpful assistant.<lim_end>

<lim_start>user

Two sides of scalene $\triangle ABC$ measure 4 centimeters and 7 centimeters. How many different whole centimeter lengths are possible for the third side?

Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>

<lim_start>assistant

723

Prompt Example for Algebra

<lim_start>system

You are a helpful assistant.<lim_end>

<lim_start>user

Six aluminum cans can be recycled to make a new can. How many new cans can eventually be made from 200 aluminum cans? (Remember that the first new cans that are made can then be recycled into even newer cans!)

Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>

<lim_start>assistant

724

Prompt Example for Calculus

<lim_start>system

You are a helpful assistant.<lim_end>

<lim_start>user

For how many values of x in $[0, 2\pi]$ is $\cos^{-1}(\cos 4x) = \sin^{-1}(\sin x)$?

Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>

<lim_start>assistant

725

Prompt Example for Counting and Probability

<lim_start>system

You are a helpful assistant.<lim_end>

<lim_start>user

7 white balls and k black balls are placed into a bin. Two balls are drawn at random. The probability that one ball is white and the other is black is $\frac{35}{66}$. Find the smallest possible value of k .

Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>

<lim_start>assistant

726

Prompt Example for AMC23-Q19

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
How many digits are in the base-ten representation of ' $8^5 \cdot 5^{10} \cdot 15^5$ '?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

727

Prompt Example for AIME24-Q1

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Every morning Aya goes for a 9 -kilometer-long walk and stops at a coffee shop afterwards. When she walks at a constant speed of s kilometers per hour, the walk takes her 4 hours, including t minutes spent in the coffee shop. When she walks $s+2$ kilometers per hour, the walk takes her 2 hours and 24 minutes, including t minutes spent in the coffee shop. Suppose Aya walks at $s+\frac{1}{2}$ kilometers per hour. Find the number of minutes the walk takes her, including the t minutes spent in the coffee shop.
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

728

Prompt Example for OlympiadBench-Q1631

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
For a positive integer a , define a sequence of integers $x_1, x_2, \dots$ by letting $x_1 = a$ and $x_{n+1} = 2x_n + 1$ for $n \geq 1$. Let $y_n = 2^{x_n} - 1$. Determine the largest possible k such that, for some positive integer a , the numbers $y_1, \dots, y_k$ are all prime.
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

729